MEDFORD-RNASeq: A human and machine readable markup language to facilitate FAIR coral metadata

John Freeman, Jill Ashey, Polina Shpilker, Hailey McKelvie, Hollie M. Putnam,
Jane Greenberg, Lenore J. Cowen, Alva Couch, Noah Daniels
Tufts University, Drexel University, University of Rhode Island

Coral Researchers collect diverse data worldwide

Coral researchers collect genomic data using technologies such as RNASeq, but also other kinds of data, including still images, video, and physical measurements [1,2].

These data associate with even more diverse metadata

Any data collected by a coral researcher associates with a wide variety of metadata: geographical location, temperature and other environmental conditions, species - but also, the technologies used to collect the data, affiliations and identities of the researchers, publication venues, and the location of the data.

Some repositories require coral metadata to be deposited

The National Science Foundation funds the Biological & Chemical Oceanography Data Management Office (BCO-DMO) [3,4], which requires researchers to deposit metadata about coral datasets resulting from NSF-funded research. BCO-DMO currently comprises 1,260 projects, 9,706 datasets, and 2,449 publications [5].

These forms are typically labor-intensive

Dataset Metadata Submission Form

For full instructions, see our "See How to Get Started" page (http://www.bco-dmo.org/how-get-started)
Please send your completed form or questions to info@bco-dmo.org.

Dataset Name *[Preferred short name for the dataset (preferably 30 characters or less)]*

Dataset Description: *[Brief sentence describing these data (preferably 60 characters or less)]*

Originating PI name and contact information:

Name:

Email:

Phone:

Mailing Address:

MEDFORD is structured for both humans and machines to easily read and write

MEDFORD is designed to be both structured and human readable. Coral researchers, technicians, and non-programmers can easily enter and validate metadata. MEDFORD eschews complex structure, favoring a flat file format. A MEDFORD metadata file can be packaged with raw data files or contain links or external references to heterogeneous raw data files.

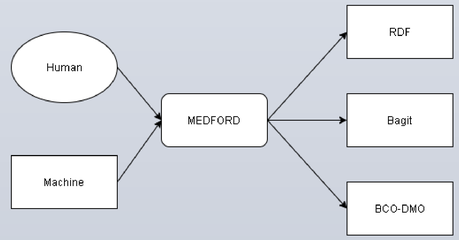
```
@Paper Whole Transcriptome Analysis of the Coral Acropora millepora Reveals Comp
@Paper-DOI 10.1111/j.1365-294X.2012.05554.x
@Paper-PMID 22490231
@Paper-Link https://pubmed.ncbi.nlm.nih.gov/22490231/
@Paper-Link https://onlinelibrary.wiley.com/doi/abs/10.1111/j.1365-294X.2012.055

#Dates are formatted YYYY-DD-MMM:HH:MM:SS.ss where the ss is optional
@Date 2012-10-APR:00:00:00
@Date-Type Published Date

#This example has no Species ID or reef
@Species  Acropora millepora
@Species-Loc Orpheus Island, Queensland, Australia
@Species-Notes Cultivated and settled under 380 ppm pCO2 pHNBS 8.16 +- 0.01, 27.

#All software will be marked by their name then optional subfields for Notes, ve
@Software NanoDrop ND-10000
@Software-Type RNA quantity and quantity
@Software-Note denaturing gel electrophoresis using standard methods. RNA sample
@Software-Version ND-1000

@Software Rest©
@Software-Type Relative expression


#Data and external links will be marked by their names then subfields for URI, R
@Data Supplementary methods
@Data-URI https://onlinelibrary-wiley-com.ezproxy.library.tufts.edu/action/downl
@Data-Notes Supporting info item. This contains all of the figures, tables, and
```

MEDFORD combines ease of data entry with ease of parsing

We present MEDFORD (MEtaData Format for Open Reef Data), initially focused on metadata relating to RNASeq studies of coral holobionts. MEDFORD can cope with the complexity of coral research metadata, while supporting FAIR use [6].

MEDFORD targets existing and future datasets

MEDFORD's parser targets a variety of existing data repositories, including BCO-DMO, BagIt [7], and RDF (Resource Description Framework) [8]. The lightweight MEDFORD parser enables data validation, so that researchers can be warned about data-entry errors or missing information. Entry can be done by human or machine.



ACKNOWLEDGEMENTS

Supported in part by the National Science Foundation under grants 1939795 (to HP), 1940239 (to JG), and 1939263 (to LC).

REFERENCES

[1] Liew, Y.J., Aranda, M. and Voolstra, C.R., 2016. Reefgenomics.Org - a repository for marine genomics data. *Database*, *2016*. baw152.

[2] Madin, J.S., Anderson, K.D., Andreasen, M.H., Bridge, T.C., Cairns, S.D., Connolly, S.R., Darling, E.S., Diaz, M., Falster, D.S., Franklin, E.C. and Gates, R.D., 2016. The Coral Trait Database, a curated database of trait information for coral species from the global oceans. *Scientific Data*, *3*(1), pp.1-22.

[3] Groman, R.C., Allison, M.D., Chandler, C.L., Copley, N.J., Gegg, S.R., Glover, D.M., Kinkade, D.B., Rauch, S.M., Shepherd, A. and Wiebe, P.H., 2014. Progress in Data Discovery and Access: The Next Steps.

[4] Allen, J.M., 1992. Woods Hole.

[5] https://www.bco-dmo.org/ Accessed December 6, 2020

[6] Wilkinson, M.D., Dumontier, M., Aalbersberg, I.J., Appleton, G., Axton, M., Baak, A., Blomberg, N., Boiten, J.W., da Silva Santos, L.B., Bourne, P.E. and Bouwman, J., 2016. The FAIR Guiding Principles for scientific data management and stewardship. *Scientific data*, *3*(1), pp.1-9.

[7] Allen, J.M., 1992. Woods Hole.

Littman, J., L. Madden, and B. Vargas. "The BagIt File Packaging Format (V0. 97) draft-kunze-bagit-07. txt." (2012).

[8] Grobe, M., 2009, October. Rdf, jena, sparql and the 'semantic web'. In Proceedings of the 37th annual ACM SIGUCCS fall conference: communication and collaboration (pp. 131-138).

RESEARCH POSTER PRESENTATION DESIGN © 2015
www.PosterPresentations.com